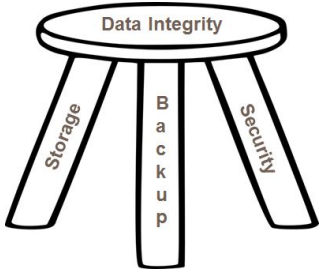
Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 4

Data Storage and Security



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Learning Objectives

1. Identify storage options and strategies for your research data that will enable long-term access in spite of potential media failures or obsolescence
2. Identify a backup strategy that will allow you to restore your data in the short- or longer-term in the event that it is lost or destroyed
3. Describe options for restricting access to your research data
4. Describe mechanisms for securing hardware and software systems
5. Describe mechanisms for protecting the integrity of your data

Storage, Backup & Security

- They are fundamental and interrelated components of a data management strategy
- Together they ensure the ongoing integrity of research data
- Project planning and data management must account for all three

Storage

- Paper alone (lab notebooks) cannot provide the levels of security now available for digital media
- Digital media to which you save your data files and software (optical, magnetic)
- All storage media are vulnerable to risk
- Evaluate and refresh regularly

Storage

- Internal, local hard drives
- Networked storage
- External storage devices
- Remote storage services (The Cloud)
- Physical storage

** Labeling is critical for any external storage device

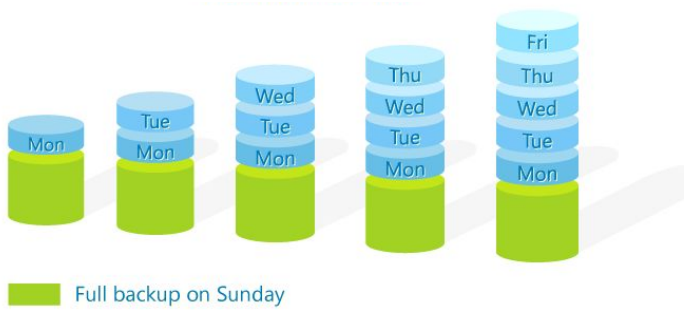
Backup

- Allows you to *restore* your data if it is lost
  - Hardware or software malfunction
  - Environmental disaster (fire, flood)
  - Theft
  - Unauthorized access

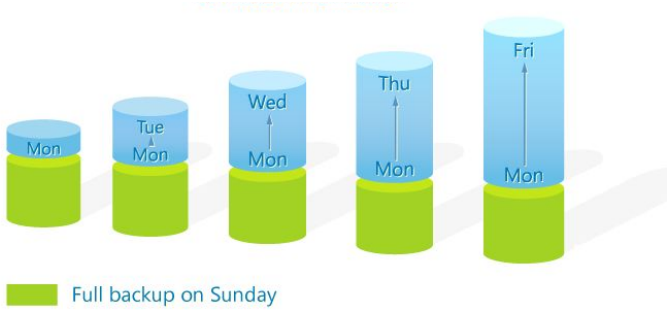
** Does your operating system and/or network administrator already have existing backup protocols and tools in place?

Backup

- Backup Procedure
  - Full
  - Differential
  - Incremental



- Backup Frequency
  - Daily
  - Weekly
  - Monthly

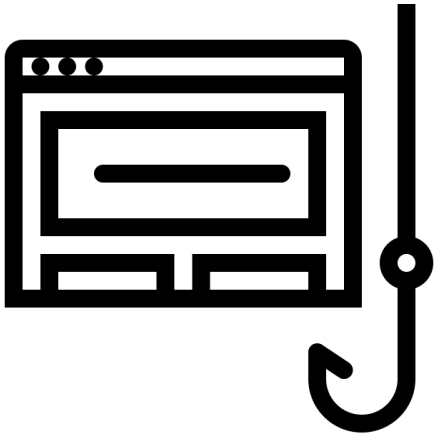


Backup

Backup Rule of Three:

1. Three copies of your data
2. Two different storage types
3. One copy of the data offsite

Data Security



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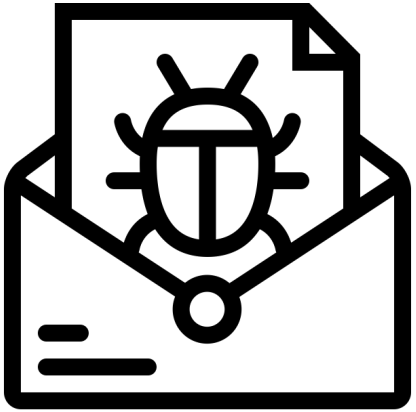

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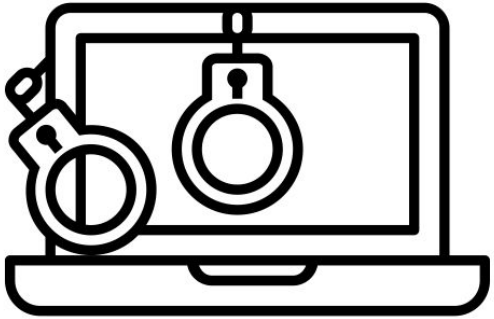

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Phishing: obtain sensitive information by masquerading as a trustworthy entity

Malware: software used to disrupt; or gain access and information

Ransomware: malware that attacks and encrypts a computer/system

Levels of Security

- Safeguard your data from malicious or accidental access to your system
  - Limit the access to your data
  - Protect the integrity of your data at the file level
  - Use software that can protect your data and its integrity
  - Protect the hardware and software systems you use

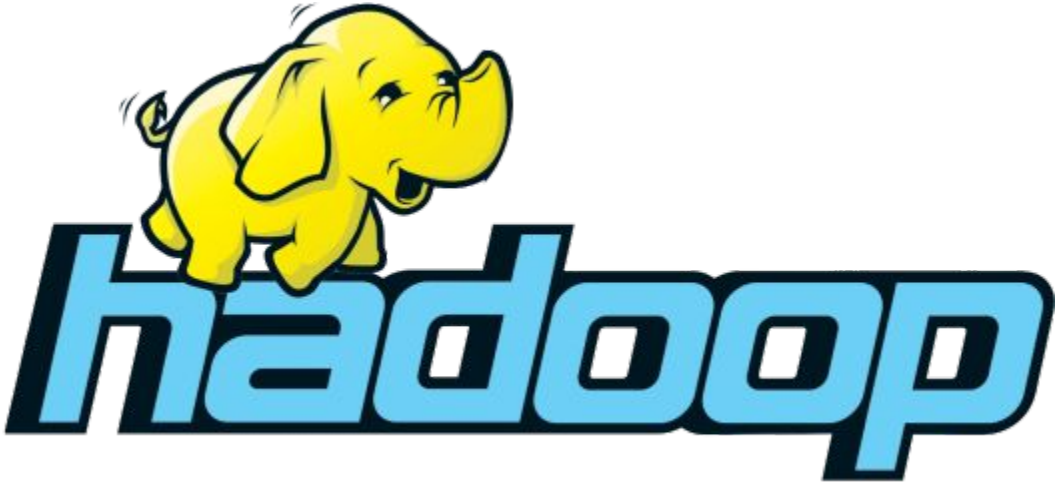
** Does your network administrator have security services that meet your needs?

Electronic Lab Notebook

- ELNs offer many benefits:
  - easier to search upon
  - simplify data copying and backups
  - support collaboration amongst users
  - can have fine-grained access controls
  - can be more secure than their paper counterparts
  - allow direct incorporation of data from instruments

Research Electronic Data Capture

REDCap is a free, secure, web-based application designed to support data capture for research studies

Apache Hadoop

Hadoop is an open-source software framework used for distributed storage and processing of very large data sets